

This is just one of the versions of the exam papers. Your exam question set might have different questions from this one.
This sample solution is for personal use, so please do not share this with others.

Questions 1 to 4 are about the predicate $x < 4 \wedge b[x] \neq 3 \rightarrow b[x] > b[0]$.

1. Which one of the following is syntactically equivalent to $x < 4 \wedge b[x] \neq 3 \rightarrow b[x] > b[0]$? (3 pts)

- a. $(x < 4 \wedge b[x] \neq 3) \rightarrow b[x] > b[0]$
- b. $x < 4 \wedge (b[x] \neq 3 \rightarrow b[x] > b[0])$
- c. $\forall x. x < 4 \wedge b[x] \neq 3 \rightarrow b[x] > b[0]$
- d. None of the above.

2. Which one of the following is semantically equivalent to $x < 4 \wedge b[x] \neq 3 \rightarrow b[x] > b[0]$? (3 pts)

- a. $b[x] \neq 3 \wedge x < 4 \rightarrow b[x] > b[0]$
- b. $x < 4 \wedge b[x] = 3 \vee b[x] > b[0]$
- c. $b[x] \neq 3 \rightarrow b[x] > b[0] \wedge x < 4$
- d. None of the above.

3. Which one of the following states is proper for $x < 4 \wedge b[x] \neq 3 \rightarrow b[x] > b[0]$? (3 pts)

- a. $\{x = 5, b = (1, 3, 5)\}$
- b. $\{b = \beta\}$, where $\beta = (1, 4, 3, 2)$
- c. $\{x = 3, b = 3\}$
- d. None of the above.

4. Present a well-form state that satisfies $x < 4 \wedge b[x] \neq 3 \rightarrow b[x] > b[0]$. (6 pts)

A possible state is $\{x = 1, b = (4, 3)\}$

Questions 5 and 6 are about expression $\text{if } x < 5/x \text{ then } y \text{ else } z \text{ fi} = 0$, here } x, y, z \text{ are primitive variables.}

5. Which one of the following about the expression $\text{if } x < 5/x \text{ then } y \text{ else } z \text{ fi} = 0$ is true? (3 pts)

- a. State $\{x = 1, y = T, z = 2\}$ is proper for this expression.
- b. If state $\sigma \models x < 0$, then $\sigma(\text{if } x < 5/x \text{ then } y \text{ else } z \text{ fi} = 0) = \perp_e$.
- c. Variables x, y and z in this expression must have the same data type.
- d. None of the above.

6. Evaluate expression $\text{if } x < 5/x \text{ then } y \text{ else } z \text{ fi} = 0$ in state $\tau = \{x = 2, y = 6, z = 0\}$. Show your work. (10 pts)

$$\begin{aligned}
 \tau(\text{if } x < 5/x \text{ then } y \text{ else } z \text{ fi} = 0) &= \tau(\text{if } F \text{ then } y \text{ else } z \text{ fi} = 0) \quad \# \text{ Note that, } 5/2 = 2 \\
 &= \tau(z = 0) \\
 &= \tau(0 = 0) \\
 &= T
 \end{aligned}$$

Questions 7 to 9 are about state $\sigma = \{p = T, q = F\}$.

7. Which one of the following statements about updating state σ is true? (3 pts)
- $\sigma[p \mapsto p \vee q] = \{p = T, q = F\}$
 - $\sigma[p \mapsto T] = \sigma$
 - $\sigma \cup \{p = F\} = \{p = F, q = F\}$
 - None of the above.
8. Let state $\tau = \sigma[p \mapsto F]$. Which one of the following is correct about the collection $\{\sigma, \tau\}$? (3 pts)
- $\{\sigma, \tau\} \models p$ or $\{\sigma, \tau\} \models q$
 - $\{\sigma, \tau\} \models p$ or $\{\sigma, \tau\} \models \neg p$
 - $\{\sigma, \tau\} \models q$ or $\{\sigma, \tau\} \models \neg q$
 - None of the above.
9. Let state $\gamma = \sigma[x \mapsto 1][p \mapsto F]$, find the state $\sigma[y \mapsto \gamma(x)][p \mapsto \sigma(\neg p \vee q)]$. Show your work. (12 pts)

$$\gamma = \{p = F, q = F, x = 1\}$$

$$\begin{aligned} \sigma[y \mapsto \gamma(x)][p \mapsto \sigma(\neg p \vee q)] &= \{p = T, q = F\}[y \mapsto \gamma(x)][p \mapsto \sigma(\neg p \vee q)] \\ &= \{p = T, q = F, y = 1\}[p \mapsto \sigma(\neg p \vee q)] \\ &= \{p = T, q = F, y = 1\}[p \mapsto (F \vee F)] \\ &= \{p = F, q = F, y = 1\} \end{aligned}$$

Questions 10 to 12 are about a predicate function $Max(b, n)$.

10. Let b be an integer array and $size(b) > n > 0$. Define predicate function $Max(b, n)$ such that it returns true if and only if $b[n]$ is greater than every other number in array b . (10 pts)

$$Max(b, n) \equiv size(b) > n > 0 \wedge \forall i. 0 \leq i < size(b) \wedge i \neq n \rightarrow b[n] > b[i]$$

11. Which one of the following collections of states satisfies $Max(b, 3)$? (3 pts)
- $\{b = (4)\}$
 - $\{b = (1, 2, 3, 3), n = 3\}, \{b = (2, 3, 4, 4), n = 3\}$
 - $\{b = (6, 4, 3, 5)\}, \{b = (1, 3, 2, 4)\}$
 - None of the above.
12. Let $K(b, n) \equiv size(b) > n > 0 \wedge \forall x. x \neq n \wedge 0 < x \leq size(b) \rightarrow b[x] < b[n]$. Which one of the following predicates is valid? (3 pts)
- $K(b, n) = Max(b, n)$
 - $K(b, n) \rightarrow \neg Max(b, n)$
 - $K(b, n) \vee Max(b, n)$
 - None of the above.

Questions 13 and 14 are about the following statement.

$S \equiv y := x + 1; x := 2; W$, where $W \equiv \text{while } y < 4 \text{ do } y := y + x \text{ od}$

13. What is the configuration $\langle S, \{x = 3\} \rangle \rightarrow^2$? (3 pts)
- $\langle y := x + 1; W, \{x = 2\} \rangle$
 - $\langle x := 2; W, \{x = 3, y = 4\} \rangle$
 - $\langle W, \{x = 2, y = 4\} \rangle$
 - None of the above.

14. Calculate $M(S, \sigma)$, where $\sigma(x) = 2$ and $\sigma(y) = 6$. Show your work. (12 pts)

$$\begin{aligned}
 M(S, \sigma) &= M(x := 2; W, \sigma[y \mapsto 3]) \\
 &= M(W, \sigma[y \mapsto 3][x \mapsto 2]) \\
 &= M(y := y + x; W, \sigma[y \mapsto 3][x \mapsto 2]) \\
 &= M(W, \sigma[y \mapsto 5][x \mapsto 2]) \\
 &= \{\sigma[y \mapsto 5][x \mapsto 2]\}
 \end{aligned}$$

Questions 15 to 21 are about Hoare triples.

15. Which one of the following is a valid triple under partial correctness?
- $\{x < 0\} x := x * x \{x \leq 0\}$
 - $\{x = 0\} x := x * x \{x \leq 0\}$
 - $\{x > 0\} x := x * x \{x \leq 0\}$
 - None of the above.
16. Let $\sigma \not\models_{tot} \{p\} S \{q\}$. Which one of the following must be true? (3 pts)
- $\sigma \models p$ and $M(S, \sigma) \not\models q$
 - $\sigma \not\models p$ or $M(S, \sigma) \perp \not\models q$
 - $\sigma \not\models p$ and $M(S, \sigma) \not\models q$
 - None of the above.
17. Let $\not\models_{tot} \{p\} S \{q\}$. Which one of the following must be true? (3 pts)
- There exists some state σ such that $\sigma \models p$ and $M(S, \sigma) \not\models q$.
 - There exists some state σ such that $\sigma \not\models p$.
 - For each proper state σ , it is the case that $M(S, \sigma) \not\models q$.
 - None of the above.
18. Let $\models \{p\} S \{q\}$ but $\not\models_{tot} \{p\} S \{q\}$. Which one of the following must be true? (3 pts)
- There exists some σ such that $\perp \in M(S, \sigma)$.
 - There exists some σ such that $M(S, \sigma) \models q$.
 - There exists some σ such that $M(S, \sigma) \perp \not\models q$.
 - None of the above.
19. Let $\{x > 1\} S \{x > 1\}$ be valid under total correctness. Which one of the following triples must be totally valid? (3 pts)
- $\{x > 1\} S \{x > 2\}$
 - $\{x > 0\} S \{x > 1\}$
 - $\{x > 1\} S; S \{x > 1\}$
 - None of the above.

20. Let $\models_{tot} \{x > 1\} S \{x \leq 1\}$. Which one of the following must be true? (3 pts)
- $M(S, \{x = 2\}) \models x \leq 2$
 - $M(S, \{x = 1\}) \models x \leq 1$
 - $M(S, \{x = 1\}) \models x > 1$
 - None of the above.
21. Given that $\models_{tot} \{T\} S \{x > 1\}$ and let σ be a proper state for S . Which one of the following must be true? (3 pts)
- S is deterministic.
 - $M(S, \sigma) \models x < 100$
 - $\sigma \models \{x < 100\} S \{x > 1\}$
 - None of the above.

Questions 22 to 24 are about the following nondeterministic statement

$$NF \equiv \text{if } x \geq 3 \rightarrow b[x] := b[b[x]] \square x < 1 \rightarrow b[x] := 1 / b[b[x]] \text{ fi}$$

22. Let σ be a proper state for NF . Which one of the following about $M(NF, \sigma)$ is true? (3 pts)
- It is possible that $\perp_d \in M(NF, \sigma)$.
 - It is possible that $M(NF, \sigma) \models (x = 2)$.
 - It is possible that $M(NF, \sigma) = \{\perp_e\}$.
 - None of the above.
23. Let τ be a state proper for NF , and $\tau(b) = (-1, 2, 0, 2)$. Which one of the following statements about the execution of NF in state τ is true? (3 pts)
- If $\tau(x) = 1$, then a “division by 0” error will occur in the evaluation of $\langle NF, \tau \rangle$.
 - If $\tau(x) = 2$, then an “all guards are False” error will occur in the evaluation of $\langle NF, \tau \rangle$.
 - If $\tau(x) = 3$, then an “array index out of bound” error will occur in the evaluation of $\langle NF, \tau \rangle$.
 - None of the above
24. Present a well-formed state σ so that $M(NF, \sigma)$ consists of exactly one well-formed state. (6 pts)

A possible state is $\sigma = \{x = 3, b = (1, 1, 1, 1)\}$